Benchmark residual invariants in compact S^3 cosmology

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We formulate a benchmark-grade, first-principles framework for compact S^3 cosmology. The useful object is not a verbal explanation of the dark sector, but a reproducible map from invariant inputs to data likelihoods. Starting from $(M, g_{\mu\nu}, \Sigma, U^\mu, T^{\rm vis}_{\mu\nu})$, with $M \simeq I \times S^3$, we define the residual stress tensor, the proper-time background reconstruction, the S^3 spectral covariance kernels, the Noether-II and quasi-local constraints, and the data map needed for CMB, BAO, supernova, growth, lensing, and topology tests. We then add the standard that makes the construction useful to other groups: a benchmark manifest, units and priors for each invariant parameter, a transfer-operator ledger, validation tests, posterior predictive residuals, evidence comparisons, and public reference outputs. This submission defines the benchmark-specification version of the framework; numerical reference spectra, kernels, and likelihood tables are deliberately separated into a subsequent validation release. The result is not a microphysical dark-sector model and does not assert that topology alone explains cosmic acceleration. It is a strict invariant-to-data benchmark against which compact-curvature and residual-stress hypotheses can be tested, reproduced, and falsified.

I. INTRODUCTION

The problem is not to attach a new name to the dark sector. The problem is to state the object which a successful dark-sector theory must match. In a relativistic theory that object must be tensorial, conserved under the appropriate hypotheses, and reconstructible from observables without choosing a coordinate gauge; the conventions are those of standard general relativity and relativistic cosmology [1–5].

We use the following first-principles data:

$$(M, g_{\mu\nu}, \Sigma, U^\mu, T^{\rm vis}_{\mu\nu}), \qquad M \simeq I \times S^3, \qquad U^\mu U_\mu = -c^2. \tag{1}$$

Here $g_{\mu\nu}$ is the spacetime metric, Σ is a compact Cauchy slice diffeomorphic to S^3, U^μ is the comoving timelike congruence, and $T^{\rm vis}_{\mu\nu}$ is the specified visible stress tensor, including baryons, photons, neutrinos, and any other sector one chooses to count as visible.

The central definition is

$$T^{\mathcal{D}}_{\mu\nu} := \frac{c^4}{8\pi G} G_{\mu\nu} - T^{\rm vis}_{\mu\nu}. \tag{2}$$

It is a residual target tensor, not a microscopic model. It is "complete" only in this restricted sense: once $g_{\mu\nu}$ and $T^{\rm vis}_{\mu\nu}$ are specified, Eq. (2) determines the dark-sector target. It does not determine whether the target is produced by particles, fields, modified gravity written in Einstein form, interactions, or effective coarse graining.

The present version adds the part that makes the framework predictive. It gives a network of invariants, specifies which of them lead to observables, and states the likelihood structure needed for CMB, BAO, supernova, lensing, growth, topology, and quasi-local tests. We keep three rules throughout. First, no quantity is called invariant merely because it is useful. Scalars, spectra, topological numbers, covariants, and gauge-redundant objects are separated. Second, any claim that depends on a bridge assumption is marked conditional. Third, numerical claims are not promoted to predictions until a transfer function, likelihood, and data covariance have been specified.

II. GEOMETRY AND PROPER TIME

Let

$$\mathrm{d}s^2 = -N^2(t)c^2\mathrm{d}t^2 + a^2(t)R_0^2\mathrm{d}\Omega_3^2, \tag{3}$$

where $\mathrm{d}\Omega_3^2$ is the unit three-sphere metric. The physical curvature radius is

$$A(t) := a(t)R_0. \tag{4}$$

The coordinate t has no invariant meaning unless N is fixed. Therefore we use

$$f' := U^\mu \nabla_\mu f = \frac{1}{N}\frac{\mathrm{d}f}{\mathrm{d}t}, \qquad H := \frac{a'}{a} = \frac{\dot{a}}{Na}. \tag{5}$$

A dot denotes $\mathrm{d}/\mathrm{d}t$; a prime denotes differentiation with respect to proper time τ.[1]

For S^3_A,

$$V = 2\pi^2 A^3, \qquad \mathcal{R}^{(3)} = \frac{6}{A^2}, \qquad \mathrm{diam}(S^3_A) = \pi A. \tag{6}$$

The curvature parameter is

$$\Omega_K = -\frac{c^2}{A_0^2 H_0^2}, \qquad A_0 = \frac{c}{H_0\sqrt{-\Omega_K}} \quad (\Omega_K < 0). \tag{7}$$

Thus the flat limit is not a topological limit inside S^3; it is the geometric limit $A_0 H_0/c \to \infty$.

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[1] The dot is used only for coordinate-time differentiation before lapse fixing; all dynamical background equations below use the prime/proper-time convention.

III. RESIDUAL STRESS-ENERGY

Proposition 1 (Covariance and conservation)**.** *The tensor $T^{\mathcal{D}}_{\mu\nu}$ defined in Eq. (2) is a symmetric rank-two tensor. If the visible stress tensor is separately conserved,*

$$\nabla_\mu T^{\mathrm{vis}\,\mu\nu} = 0, \tag{8}$$

then

$$\nabla_\mu T^{\mathcal{D}\,\mu\nu} = 0. \tag{9}$$

Proof. The Einstein tensor and $T^{\mathrm{vis}}_{\mu\nu}$ are tensors, so their linear combination is a tensor. Taking a divergence of Eq. (2) gives

$$\nabla_\mu T^{\mathcal{D}\,\mu\nu} = \frac{c^4}{8\pi G}\nabla_\mu G^{\mu\nu} - \nabla_\mu T^{\mathrm{vis}\,\mu\nu}. \tag{10}$$

The first term vanishes by the contracted Bianchi identity; the second vanishes by assumption. □

If visible and dark sectors exchange energy-momentum, the correct statement is instead

$$\nabla_\mu T^{\mathrm{vis}\,\mu\nu} = J^\nu, \qquad \nabla_\mu T^{\mathcal{D}\,\mu\nu} = -J^\nu, \tag{11}$$

so that the sum is conserved. A reconstructed nonzero divergence is therefore not a falsification of S^3 topology. It is evidence for an inconsistent reconstruction, a non-conserved visible sector, or a missing interaction current.

For the FLRW background define

$$\epsilon_{\mathrm{crit}} := \frac{3c^2H_0^2}{8\pi G}, \qquad E(a) := \frac{H(a)}{H_0}. \tag{12}$$

The Friedmann equations give

$$\frac{\epsilon_{\mathrm{tot}}}{\epsilon_{\mathrm{crit}}} = E^2 - \Omega_K a^{-2}, \tag{13}$$

and

$$\frac{p_{\mathrm{tot}}}{\epsilon_{\mathrm{crit}}} = -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2 - \Omega_K a^{-2}\right]. \tag{14}$$

The proper-time acceleration equation and deceleration parameter are

$$\frac{a''}{a} = -\frac{4\pi G}{3c^2}(\epsilon_{\mathrm{tot}} + 3p_{\mathrm{tot}}), \qquad q = -\frac{aa''}{(a')^2}. \tag{15}$$

Let

$$\Omega_{\mathrm{vis}}(a) = \Omega_b a^{-3} + \Omega_\gamma a^{-4} + \Omega_\nu(a). \tag{16}$$

Then

$$\frac{\epsilon_{\mathcal{D}}}{\epsilon_{\mathrm{crit}}} = E^2 - \Omega_K a^{-2} - \Omega_{\mathrm{vis}}(a), \tag{17}$$

and

$$\frac{p_{\mathcal{D}}}{\epsilon_{\mathrm{crit}}} = -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2 - \Omega_K a^{-2}\right] - \frac{1}{3}\Omega_\gamma a^{-4} - P_\nu(a), \tag{18}$$

where $P_\nu(a) = p_\nu(a)/\epsilon_{\mathrm{crit}}$. These formulae are background targets. They do not split $T^{\mathcal{D}}_{\mu\nu}$ uniquely into clustering dark matter and smooth dark energy; that split requires perturbative observables.

IV. S^3 SPECTRA AND COVARIANCE KERNELS

The scalar Laplacian on S^3_A has eigenvalues

$$-\Delta Y_{n\ell m} = \frac{n(n+2)}{A^2}Y_{n\ell m}, \qquad n = 0, 1, 2, \ldots, \tag{19}$$

with degeneracy $(n+1)^2$ in the standard scalar-harmonic normalization on three-spheres [6–10]. The first nonzero scalar gap is

$$\lambda_1 = \frac{3}{A^2}. \tag{20}$$

The addition kernel is

$$\begin{aligned}\Xi_{n,A}(x,y) &= \sum_{\ell,m} Y_{n\ell m}(x)Y^*_{n\ell m}(y),\\ \Xi_{n,A}(0) &= \frac{(n+1)^2}{2\pi^2A^3}.\end{aligned} \tag{21}$$

For a statistically isotropic scalar Gaussian field,

$$C(x,y) = \sum_{n=0}^{\infty} A_n\,\Xi_{n,A}(x,y) \tag{22}$$

is a covariance only if

$$A_n \geq 0, \qquad \sum_{n=0}^{\infty} A_n\,\Xi_{n,A}(0) < \infty. \tag{23}$$

Stronger decay controls differentiability. Equation (23) is a sharp admissibility condition: without it the coefficients are not a covariance spectrum.

CMB and large-scale-structure observables enter through transfer functions, as in the standard Boltzmann-code approach [25–27]. Schematically, for CMB fields $X, Y \in \{T, E, B, \phi\}$,

$$C_\ell^{XY}(\theta) = \sum_{n\in\mathcal{N}_\ell} A_n(\theta)\,\Delta^X_{\ell n}(\theta)\,\Delta^Y_{\ell n}(\theta), \tag{24}$$

where $\mathcal{N}_\ell$ is the allowed discrete mode set and $\Delta^X_{\ell n}$ are the closed-FLRW transfer functions with the chosen normalization. Equation (24) is the predictive replacement for any verbal statement about low-ℓ anomalies. Closed-FLRW hyperspherical transfer functions are already implemented in non-flat Boltzmann technology; the compact benchmark adds the discrete sum over allowed n and the covariance-admissibility condition in Eq. (23). A deficit, oscillation, alignment, or parity statistic is a derived likelihood statement, not a prior conclusion.

Vector and tensor perturbations require their own operator ledger. Scalar eigenvalues cannot be re-used for transverse vectors or transverse-traceless tensors without the correct curvature shifts, constraints, and gauge conditions.

V. NOETHER-II, QUASI-LOCAL, AND GAUGE CONSTRAINTS

Diffeomorphism invariance is governed by Noether's second theorem and the covariant phase-space identities, not by ordinary global Noether-I charges [1, 11]. The quasi-local examples below use the Brown–York and Misner–Sharp constructions [12, 13]; the quantum-clock caution uses the Page–Wootters benchmark [15]. On a compact Cauchy slice without boundary,

$$\partial\Sigma = \varnothing, \tag{25}$$

the diffeomorphism Hamiltonian is a constraint for pure gauge deformations. Physical energy diagnostics arise from subregions $D \subset \Sigma$ with $\partial D \neq \varnothing$, from matter clocks, or from asymptotic/boundary structures when such structures exist.

For a geodesic two-sphere of areal radius R_A in a time-symmetric round S^3 slice, the Brown–York energy with flat reference subtraction is [12, 14]

$$E_{\rm BY}(R_A) = \frac{c^4 R_A}{G}\left[1 - \sqrt{1 - \frac{R_A^2}{A^2}}\right], \qquad 0 \le R_A \le A. \tag{26}$$

For $R_A \ll A$,

$$E_{\rm BY}(R_A) = \frac{c^4 R_A^3}{2GA^2} + \mathcal{O}(R_A^5/A^4). \tag{27}$$

The leading term has volume scaling. A Newtonian interpretation requires the constraint equations and a specified reference subtraction; it is not automatic.

The Misner–Sharp energy in FLRW is

$$E_{\rm MS}(R_A) = \frac{4\pi}{3} R_A^3 \epsilon_{\rm tot}. \tag{28}$$

Equations (26) and (28) are different quasi-local diagnostics: $E_{\rm BY}$ uses an embedding/reference subtraction, while $E_{\rm MS}$ is the local spherical FLRW mass function. They should not be identified unless the geometric setting and subtraction convention justify it.

For a gauge field on a compact slice, the Poisson equation has a zero-mode constraint:

$$\int_\Sigma \rho \, {\rm d}V = 0. \tag{29}$$

Equivalently, the Green function must be zero-mode subtracted:

$$-\Delta G_0(x,y) = \delta_y(x) - \frac{1}{V(\Sigma)}. \tag{30}$$

This is a global solvability condition, not a claim that every local flux integral vanishes.

VI. PREDICTIVE INVARIANT-TO-DATA PIPELINE

A model becomes experimentally useful only after it supplies a map

$$\theta \longmapsto m(\theta) \tag{31}$$

from invariant parameters to a data vector. A minimal data vector is

$$\begin{aligned} d = \{&C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_\ell^{\phi\phi}, D_M(z_i)/r_d, D_H(z_i)/r_d, \\ &\mu_{\rm SN}(z_j), H(z_k), f\sigma_8(z_l), \xi_\pm(\vartheta_m), \mathcal{T}_{\rm top}\}. \end{aligned} \tag{32}$$

Here $\mathcal{T}_{\rm top}$ denotes any topology statistic, such as a circle-search statistic for a multiply connected quotient S^3/Γ or a geodesic-wrap statistic for simply connected S^3, in the sense of compact-topology searches, if the horizon scale is large enough.

A corresponding parameter vector may be written

$$\begin{aligned} \theta = \{&A_0, H_0, \Omega_b h^2, \Omega_\gamma h^2, \Omega_\nu h^2, \\ &A_n \text{ or } P_{\mathcal{R}}(n), \alpha_{\mathcal{D}}, \eta_{\rm int}, \nu_{\rm nuis}\}. \end{aligned} \tag{33}$$

The symbols $\alpha_{\mathcal{D}}$ parameterize the residual background or perturbations, $\eta_{\rm int}$ parameterizes possible interaction currents J^μ, and $\nu_{\rm nuis}$ denotes nuisance parameters for foregrounds, calibration, galaxy bias, intrinsic alignments, and selection functions.

The likelihood is

$$-2\ln\mathcal{L}(\theta) = [d - m(\theta)]^T \mathcal{C}^{-1} [d - m(\theta)] + \ln\det\mathcal{C} + \text{const.}, \tag{34}$$

with a non-Gaussian replacement where the experiment requires it. This is the correct form of falsifiability. One does not declare a fixed curvature radius to be excluded in the abstract. One reports a posterior or Bayes factor after specifying data, priors, transfer functions, and covariance.

Criterion 1 (Transfer-grade predictive closure)**.** *A compact S^3 residual model is transfer-grade only if it supplies:*

1. *the metric and proper-time convention;*
2. *the visible stress tensor and any interaction current;*
3. *the scalar, vector, and tensor operator ledger;*
4. *admissible covariance spectra satisfying Eq.* (23)*;*
5. *transfer functions for the data vector* (32)*;*
6. *a likelihood and reproducible numerical implementation;*
7. *a falsification statement in posterior, Bayes-factor, or predictive-residual form.*

The practical implementation is a closed-FLRW Boltzmann pipeline with discrete compact mode sums. Existing non-flat Boltzmann solvers provide the hyperspherical transfer-function technology. Compact topology adds

the finite-volume mode ledger, possible quotient selection rules, admissible covariance coefficients, and topology likelihoods, with compact-topology searches providing one reference implementation of the latter step [28]. A code should therefore not be advertised as predicting an anomaly until it returns the objects in Eq. (32) with the covariance in Eq. (34).

VII. BENCHMARK STANDARD FOR SCIENTIFIC UTILITY

A framework is useful to physics only when another group can reproduce its numerical claims without guessing conventions. The benchmark object is therefore

$$\mathfrak{B} = (\mathcal{I}, \Theta, \pi, \mathcal{T}, \mathcal{N}, \mathcal{V}, \mathcal{R}), \tag{35}$$

where $\mathcal{I}$ is the invariant ledger, Θ is the parameter space with units, π is the prior measure, $\mathcal{T}$ is the transfer map from invariants to observables, $\mathcal{N}$ is the noise and covariance model, $\mathcal{V}$ is the validation suite, and $\mathcal{R}$ is the public release manifest. Omitting any one of these entries turns a prediction into an interpretation.

Definition 1 (Registered prediction)**.** *A registered prediction is a tuple*

$$P = (\theta, \pi, \mathcal{T}, d_{\rm blind}, \mathcal{C}, \mathcal{L}, \mathcal{F}) \tag{36}$$

fixed before the target data $d_{\rm blind}$ are used. The final entry $\mathcal{F}$ is the failure criterion: posterior exclusion, Bayes-factor loss, or posterior-predictive inconsistency. A fit made after seeing the data is not a prediction; it is a consistency test.

Proposition 2 (Coordinate-free reproducibility)**.** *If the entries in Eq.* (35) *are expressed through scalar invariants, tensorial projections, spectra, or quasi-local surface integrals, then the benchmark predictions are independent of coordinate charts on M and on Σ.*

Proof. The data map $m(\theta)$ uses only the invariant ledger and the experiment's projection operators. A coordinate change alters component representations of $g_{\mu\nu}$, $T^{\mathcal{D}}_{\mu\nu}$, harmonic bases, and surface embeddings, but not eigenvalues, multiplicities, addition kernels, contractions with U^μ, likelihood scalars, or surface integrals with the induced measure. Hence the numerical output is a function of geometric objects, not of labels. □

The minimal benchmark products are the following.

1. *Invariant manifest*: definitions, units, domains, priors, and exact/conditional status of all entries $\mathcal{I}_i$. This prevents hidden convention changes and non-comparable analyses.

2. *Transfer ledger*: the map from $\{\lambda_n, d_n, \Xi_{n,A}, T^{\mathcal{D}}_{\mu\nu}\}$ to C_ℓ, BAO distances, growth, lensing, and topology statistics. This prevents claims of prediction without a computable observable.

3. *Validation suite*: analytic limits, including the flat limit, large-radius limit, covariance positivity, ΛCDM residual check, and small-sphere Brown–York expansion. This prevents numerical agreement that is only accidental.

4. *Likelihood package*: data vector, covariance, masks, nuisance parameters, priors, and posterior-predictive residuals. This prevents unfalsifiable comparison with selected plots.

5. *Reference outputs*: versioned tables for spectra, kernels, background residuals, and benchmark likelihood values. This prevents irreproducible code-dependent numbers.

6. *Release manifest*: version hashes, exact datasets, compilation/runtime environment, and license. This prevents results that cannot be audited or reused.

The first numerical target is not a full explanation of CMB anomalies or cosmological tensions. The first target is more severe and more useful: given a closed-FLRW background and a compact spectral ledger, reproduce standard non-flat transfer-function limits, replace continuous long-mode integrals by the correct S^3 sums, and report the likelihood change relative to the flat and simply-connected baselines. Only after this test is passed may one discuss low-ℓ anomalies, matched circles, or residual-stress interpretations.

Quasi-local observables obey the same rule. A Brown–York or Misner–Sharp map inferred from lensing is not evidence for topology by itself. It becomes useful only if the surface, subtraction convention, mass definition, data projection, and covariance are fixed in the manifest. The output is then a geometric diagnostic that can agree with, or fail against, the residual target.

VIII. OBSERVATIONAL USES

The invariant framework produces several test classes. First, background data constrain

$$\epsilon_{\mathcal{D}}(a), \qquad p_{\mathcal{D}}(a), \qquad w_{\mathcal{D}}(a) = p_{\mathcal{D}}(a)/\epsilon_{\mathcal{D}}(a) \tag{37}$$

through Eqs. (17)–(18). BAO, supernovae, cosmic-chronometer, BBN, and CMB priors enter here through $E(a)$ [16–24], r_d, and the visible-sector densities.

Second, perturbations test the split between a clustering and a smooth residual. Distances alone cannot identify dark matter. Lensing, redshift-space distortions, CMB peaks, cluster abundance, and velocity fields are needed.

Third, the discrete S^3 spectra generate covariance predictions for low multipoles, parity statistics, alignments, and cross-correlations. These are not automatic explanations of anomalies. They are likelihood tests of Eq. (24) with admissible A_n.

Fourth, compact topology can be tested by geodesic wrap-around statistics on S^3 and by matched-circle statistics on quotients S^3/Γ. The absence of a circle signal

constrains the quotient, radius, orientation, and noise model used in that search; it is not by itself a universal exclusion of closed positive curvature.

Fifth, quasi-local diagnostics such as E_{BY} and E_{MS} can be compared with lensing mass maps and cluster observables only after specifying the slicing, reference subtraction, mass definition, and selection function. This is a difficult but well-posed extension.

IX. CONDITIONAL EXTENSIONS

Some useful mathematics is conditional rather than first-principles physics. If a hierarchy is defined by

$$F_{n+2} = F_{n+1} + F_n, \tag{38}$$

then

$$\frac{F_{n+1}}{F_n} \to \varphi = \frac{1+\sqrt{5}}{2}. \tag{39}$$

This is a Perron–Frobenius fact about a recursion. It is not a spacetime prediction unless a separate bridge connects the recursion to spectral or geometric data.

Similarly, graph spectra, Zeckendorf decompositions, Born–Oppenheimer parameters, and quantum-cosmological phases are useful only after their Hilbert spaces, operators, gauge fixing, and observables are stated. They may be compared with the residual target (2); they do not replace it.

X. DISCUSSION

The rigid content of the construction is

$$\begin{aligned}&\text{compact } S^3 + T^{\mathcal{D}}_{\mu\nu} + \text{ proper-time FLRW geometry}\\ &\quad + \text{discrete spectra} + \text{ Noether-II constraints}\\ &+ \text{compact gauge zero modes} + \text{ quasi-local diagnostics.}\end{aligned} \tag{40}$$

This is enough to define a useful research programme. It is not enough to claim a dark-sector mechanism, an explanation of the Hubble tension, or a proof of any famous conjecture. The scientific value lies in the opposite discipline: every extension must pass through invariant definitions, transfer functions, and data likelihoods.

The natural journal standard for the cosmology part is Physical Review D: the content is gravitation, cosmology, field theory, and data-facing cosmological inference. A Physical Review A version would have to be a different paper, centered on quantum information, quantum measurement, or atomic/optical analogues. The present manuscript should therefore be judged primarily as a PRD-style theory-and-phenomenology paper.

The standard for the next version is therefore not a larger list of claims but a smaller set of benchmark numbers: reference spectra, reference residual histories, reference covariance kernels, and reference likelihood values. These are the objects that let another group reproduce, refute, or improve the construction.

Appendix A: Predictive invariant ledger

This appendix records eighty-four entries. A scalar invariant is a number or scalar field. A spectral invariant is an eigenvalue, multiplicity, trace, or zeta quantity. A natural covariant is tensorial or geometric but not scalar. A conditional entry requires an additional bridge assumption.

1. $\mathcal{I}_1 = M \simeq I \times S^3$: spacetime product topology.
2. $\mathcal{I}_2 = \partial\Sigma = \varnothing$: absence of spatial boundary.
3. $\mathcal{I}_3 = \pi_1(S^3) = 0$: simple connectivity.
4. $\mathcal{I}_4 = H_0(S^3) = \mathbb{Z}$: connectedness.
5. $\mathcal{I}_5 = H_1(S^3) = 0$: no one-cycle homology.
6. $\mathcal{I}_6 = H_2(S^3) = 0$: no two-cycle homology.
7. $\mathcal{I}_7 = H_3(S^3) = \mathbb{Z}$: orientation volume class.
8. $\mathcal{I}_8 = \chi(S^3) = 0$: Euler characteristic.
9. $\mathcal{I}_9 = A(t) = a(t)R_0$: physical curvature radius.
10. $\mathcal{I}_{10} = V = 2\pi^2 A^3$: spatial volume.
11. $\mathcal{I}_{11} = \operatorname{diam} S^3_A = \pi A$: diameter.
12. $\mathcal{I}_{12} = \mathcal{R}^{(3)} = 6/A^2$: scalar curvature.
13. $\mathcal{I}_{13} = R^{(3)}_{ij} = 2h_{ij}/A^2$: Ricci tensor.
14. $\mathcal{I}_{14} = K = +1/A^2$: sectional curvature.
15. $\mathcal{I}_{15} = \Omega_K = -c^2/(A_0^2 H_0^2)$: dimensionless curvature.
16. $\mathcal{I}_{16} = A_0 H_0/c = 1/\sqrt{-\Omega_K}$: curvature radius in Hubble units.
17. $\mathcal{I}_{17} = U^\mu U_\mu = -c^2$: normalized congruence.
18. $\mathcal{I}_{18} = f' = U^\mu \nabla_\mu f$: proper-time derivative.
19. $\mathcal{I}_{19} = H = a'/a$: expansion rate.
20. $\mathcal{I}_{20} = \Theta = \nabla_\mu U^\mu = 3H$: expansion scalar.
21. $\mathcal{I}_{21} = q = -aa''/(a')^2$: deceleration scalar.
22. $\mathcal{I}_{22} = 1 + z = a_0/a$: redshift in FLRW.
23. $\mathcal{I}_{23} = E(z) = H(z)/H_0$: dimensionless expansion.
24. $\mathcal{I}_{24} = \chi(z) = \int_0^z c\,\mathrm{d}z'/H(z')$: radial comoving distance.
25. $\mathcal{I}_{25} = D_M(z) = A_0 \sin[\chi(z)/A_0]$: transverse distance for S^3.

26. $\mathcal{I}_{26} = D_H(z) = c/H(z)$: Hubble distance.
27. $\mathcal{I}_{27} = r_d$: drag-epoch sound horizon.
28. $\mathcal{I}_{28} = D_M/r_d$: BAO transverse observable.
29. $\mathcal{I}_{29} = D_H/r_d$: BAO radial observable.
30. $\mathcal{I}_{30} = G_{\mu\nu}$: Einstein tensor natural covariant.
31. $\mathcal{I}_{31} = \nabla_\mu G^{\mu\nu} = 0$: contracted Bianchi identity.
32. $\mathcal{I}_{32} = T^{\mathrm{vis}}_{\mu\nu}$: specified visible stress tensor.
33. $\mathcal{I}_{33} = T^{\mathcal{D}}_{\mu\nu}$: residual stress tensor.
34. $\mathcal{I}_{34} = \epsilon_{\mathcal{D}} = T^{\mathcal{D}}_{\mu\nu} U^\mu U^\nu / c^2$: residual energy density.
35. $\mathcal{I}_{35} = p_{\mathcal{D}} = h^{\mu\nu} T^{\mathcal{D}}_{\mu\nu}/3$: residual isotropic pressure.
36. $\mathcal{I}_{36} = w_{\mathcal{D}} = p_{\mathcal{D}}/\epsilon_{\mathcal{D}}$: residual background equation-of-state ratio.
37. $\mathcal{I}_{37} = h^\alpha{}_\mu T^{\mathcal{D}}_{\alpha\beta} U^\beta$: residual momentum flux.
38. $\mathcal{I}_{38} = \pi^{\mathcal{D}}_{\mu\nu}$: residual anisotropic stress.
39. $\mathcal{I}_{39} = T^{\mathcal{D}} = g^{\mu\nu} T^{\mathcal{D}}_{\mu\nu}$: residual trace.
40. $\mathcal{I}_{40} = \nabla_\mu T^{\mathcal{D}\,\mu\nu}$: residual divergence diagnostic.
41. $\mathcal{I}_{41} = J^\nu$: interaction current if sectors exchange energy-momentum.
42. $\mathcal{I}_{42} = -\Delta$: scalar Laplacian on S^3_A.
43. $\mathcal{I}_{43} = \lambda_n = n(n+2)/A^2$: scalar eigenvalues.
44. $\mathcal{I}_{44} = g_n = (n+1)^2$: scalar degeneracy.
45. $\mathcal{I}_{45} = \lambda_1 = 3/A^2$: scalar spectral gap.
46. $\mathcal{I}_{46} = \Xi_{n,A}(x,y)$: scalar addition kernel.
47. $\mathcal{I}_{47} = \Xi_{n,A}(0) = (n+1)^2/(2\pi^2 A^3)$: diagonal kernel.
48. $\mathcal{I}_{48} = A_n \geq 0$: covariance positivity.
49. $\mathcal{I}_{49} = \sum_n A_n \Xi_{n,A}(0) < \infty$: finite point variance.
50. $\mathcal{I}_{50} = K(s) = \operatorname{Tr} e^{-s\Delta}$: heat trace.
51. $\mathcal{I}_{51} = \zeta_\Delta(s) = \sum_{\lambda>0} \lambda^{-s}$: spectral zeta function.
52. $\mathcal{I}_{52} = \Delta^X_{\ell n}$: closed-FLRW transfer function.
53. $\mathcal{I}_{53} = C^{XY}_\ell$: angular power spectrum prediction.
54. $\mathcal{I}_{54} = P_{\mathcal{D}}(k,z)$ or its discrete analogue: residual clustering spectrum.
55. $\mathcal{I}_{55} = f\sigma_8(z)$: growth observable.
56. $\mathcal{I}_{56} = C^{\phi\phi}_\ell$: CMB lensing spectrum.
57. $\mathcal{I}_{57} = \xi_\pm(\vartheta)$: weak-lensing correlation functions.
58. $\mathcal{I}_{58} = \mathcal{T}_{\mathrm{top}}$: topology statistic.
59. $\mathcal{I}_{59} = H_\xi \approx 0$: diffeomorphism Hamiltonian constraint on closed slices.
60. $\mathcal{I}_{60} = E_{\mathrm{BY}}(R_A)$: Brown–York quasi-local energy.
61. $\mathcal{I}_{61} = E_{\mathrm{MS}}(R_A)$: Misner–Sharp energy.
62. $\mathcal{I}_{62} = \int_\Sigma \rho \, \mathrm{d}V = 0$: compact Gauss zero-mode condition.
63. $\mathcal{I}_{63} = G_0(x,y)$: zero-subtracted Green function.
64. $\mathcal{I}_{64} = Q_{\partial D}$: quasi-local surface charge on a subregion.
65. $\mathcal{I}_{65} = \mathcal{C}$: data covariance matrix in the likelihood.
66. $\mathcal{I}_{66} = \ln \mathcal{L}$: predictive likelihood scalar.
67. $\mathcal{I}_{67} = Z_{\mathcal{M}} = \int \mathcal{L}\pi \, \mathrm{d}\theta$: Bayesian evidence.
68. $\mathcal{I}_{68} = K_{12} = Z_1/Z_2$: Bayes factor between models.
69. $\mathcal{I}_{69} = r_i = d_i - m_i(\theta)$: posterior predictive residual.

Conditional mathematical extensions, not required for the core benchmark.

70. $\mathcal{I}_{70} = \varphi$: conditional Fibonacci hierarchy eigenvalue.
71. $\mathcal{I}_{71}$: conditional graph adjacency spectrum.
72. $\mathcal{I}_{72}$: conditional quantum phase or Born–Oppenheimer parameter.
73. $\mathcal{I}_{73} = (\mathcal{I}, \Theta, \pi, \mathcal{T}, \mathcal{N}, \mathcal{V}, \mathcal{R})$: benchmark manifest.
74. $\mathcal{I}_{74} = \mathcal{T}$: invariant-to-observable transfer operator.
75. $\mathcal{I}_{75} = \mathcal{N}$: experimental noise and nuisance model.
76. $\mathcal{I}_{76} = \mathcal{V}$: validation-test vector.
77. $\mathcal{I}_{77} = p(d_{\mathrm{blind}} \mid \mathcal{M})$: blind predictive density.
78. $\mathcal{I}_{78} = r^{\mathrm{pp}}_i$: posterior-predictive residual.
79. $\mathcal{I}_{79} = \Delta\chi^2_{\mathrm{pp}}$: posterior-predictive discrepancy.
80. $\mathcal{I}_{80}$: versioned reference spectrum table.
81. $\mathcal{I}_{81}$: versioned covariance-kernel table.
82. $\mathcal{I}_{82}$: versioned residual-history table.
83. $\mathcal{I}_{83}$: code-and-data release hash.
84. $\mathcal{I}_{84}$: failure criterion attached to a registered prediction.

Appendix B: Benchmark manifest

The following is the minimal manifest for a public benchmark release. It is deliberately mechanical.

1. State the geometry: A, Ω_K, a_0, lapse convention, and proper-time derivative.

2. State the visible sector: baryons, photons, neutrinos, and any interaction current. If an interaction current is absent, state $\nabla_\mu T^{\mathrm{vis}\,\mu\nu} = 0$.

3. State the residual target: $T^{\mathcal{D}}_{\mu\nu}$, $\epsilon_{\mathcal{D}}(a)$, $p_{\mathcal{D}}(a)$, and perturbative closure if used.

4. State the spectral sector: scalar, transverse-vector, or TT; operator, eigenvalues, degeneracies, and covariance coefficients.

5. State the data vector and covariance exactly as used in the likelihood.

6. Publish validation outputs for the analytic limits listed in Sec. VII.

7. Publish posterior samples, posterior-predictive residuals, evidence estimates, and the failure criterion.

This manifest is stricter than a narrative conclusion. It is also more valuable: it converts the framework into a reusable standard.

Appendix C: Quarantine rule

The framework excludes unsupported global claims from the PRD/PRA-level core. In particular, it does not contain a proof of the Riemann hypothesis, the Collatz conjecture, Fermat's last theorem, local connectivity of the Mandelbrot set, or the Poincaré conjecture. It also does not use consciousness, reincarnation, or metaphysical claims as physical assumptions. Such material may be studied elsewhere as mathematics, philosophy, or speculation, but it is not part of the transfer-grade invariant programme unless it supplies definitions, theorems, observables, and falsifiable likelihoods.

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